

10 The African penguin, *Spheniscus demersus*, lives along the coast of South Africa.

Fig. 10.1 shows two African penguins.



Fig. 10.1

- (a) One way of estimating the size of a population of African penguins is to use the mark-release-recapture method.

Suggest **three** assumptions that must be made for the mark-release-recapture method to be valid.

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..... [3]

- (b) The International Union for the Conservation of Nature (IUCN) Red List of Threatened Species has categorised the African penguin as endangered and could become extinct.

Suggest reasons why the African penguin has become endangered and could become extinct.

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- (d)** The African penguin is a member of the kingdom Animalia.

Fig. 10.2

answer = yr⁻¹ [1]

- (d)** The African penguin is a member of the kingdom Animalia.

Outline the characteristic features of the kingdom Animalia.

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